

Mehraj Hossain Mahi

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Career Objective

Motivated and research-oriented graduate seeking a Lecturer position in Information Technology and computing disciplines. With a strong background in programming, data structures, machine learning, natural language processing, and data-driven systems, I aim to contribute to undergraduate teaching, laboratory instruction, student mentoring, and academic research while fostering an engaging and research-oriented learning environment.

Academic Qualifications

Master of Science in Computer Science and Engineering (Major: Data Science) 2026 – Present

Daffodil International University (DIU), Dhaka, Bangladesh

Status: Ongoing

Research Focus: Data Science, Machine Learning, Fairness-Aware AI

Bachelor of Science in Computer Science and Engineering (CSE) Jan 2022 – Dec 2025

Daffodil International University (DIU), Dhaka, Bangladesh

CGPA: 3.85 / 4.00

Key Achievements: Dean's List Award, Hall Hero Award, Best Paper Award, Merit-Based Scholarship.

Thesis Title: *A Comprehensive Investigation of Fair Synthetic Data Generation: Unveiling the Potential of Fairness-Aware GAN*

Thesis Duration: Jan 2025 – Dec 2025

Supervisor: Dr. Md. Alamgir Kabir

Academic and Research Experience

Lab Prefect — Data Structures Laboratory 2023 – 2024

Department of Computer Science and Engineering, Daffodil International University

- Assisted students with data structures, C/C++, programming, debugging, and academic guidance.

Research Assistant Jan 2024 – August 2026

DIU NLP & ML Lab, Daffodil International University

- Conducted research in natural language processing and Mentored students in research methodology, data analysis, academic writing, and publication preparation.

Researcher Spring 2023 – August 2026

Multidisciplinary Action Research Society (MARS Lab), Daffodil International University

- Guided students in research, survey design, data analysis, literature review, and manuscript preparation.

Publications

Published Journal Articles

- M. H. Mahi**, A. R. Khan, M. H. Anik, S. R. H. Noori, A. Mahmud, and M. U. Mojumdar, "MELD: A Multilingual Ethnic Dataset of Chakma, Garo, and Marma in Bengali Script with English and Standard Bengali Translation," *Data in Brief* (Elsevier, **Scopus Q1**), vol. 61, p. 111745, 2025. DOI: [10.1016/j.dib.2025.111745](https://doi.org/10.1016/j.dib.2025.111745)
- M. H. Mahi**, M. H. Arnab, F. F. Mahib, S. H. Ayon, and A. R. Khan, "Living Heritage in Glass: A Terrarium-Based Prototype for Interpreting Panam Nagar," *Journal of Interpretation Research*, vol. 0, no. 0, 2026. DOI: [10.1177/10925872261461224](https://doi.org/10.1177/10925872261461224)
- M. H. Mahi**, A. R. Khan, and M. U. Mojumdar, "BanglaDial: A Merged and Imbalanced Text Dataset for Bengali Regional Dialect Analysis," *Data in Brief* (Elsevier, **Scopus Q1**), vol. 63, p. 112200, 2025. DOI: [10.1016/j.dib.2025.112200](https://doi.org/10.1016/j.dib.2025.112200)
- A. Sattar, M. A. M. Ridoy, **M. H. Mahi**, and A. R. Khan, "ToxiFruit: A Multi-class Image Dataset for Formalin Adulteration Detection in Fruits," *Scientific Data* (Springer Nature, **Scopus Q1, Web of Science**), 2026. DOI: [10.1038/s41597-026-08185](https://doi.org/10.1038/s41597-026-08185)
- M. H. Mahi**, A. R. Khan, Z. Hoque, and M. U. Mojumdar, "FeniVerse: A Parallel Corpus of Feni Dialect, Standard Bengali, and English," *Data in Brief* (Elsevier, **Scopus Q1**), vol. 63, p. 112250, 2025. DOI: [10.1016/j.dib.2025.112250](https://doi.org/10.1016/j.dib.2025.112250)

Manuscripts Under Review

- "Impact of Resampling Techniques on Text-Based Regional Dialect Detection," *Language Resources and Evaluation* (Springer, **Scopus Q1**), Under Review, 2025.
- "A Systematic Literature Review on Fair Data Generation," *Array* (Elsevier, **Scopus Q1**), Under Review, 2026.

Conference Proceedings

- U. Ayman, A. R. Khan, **M. H. Mahi**, T. Akter, Z. M. Koli, and M. H. I. Bijoy, "Optimizing Bengali Sentiment Analysis: A Comparison of CountVectorizer and TF-IDF with Machine Learning," in *Proc. ECCE 2025*. Available at: [IEEE Xplore](#)

2. A. R. Khan, **M. H. Mahi**, M. U. Mojumdar, N. R. Chakraborty, M. A. Kabir, and A. N. Borua, "Transformer-Based Model for Identifying Ethnic Languages in Bangladesh," in *Proc. INDIACom 2025*. Available at: [IEEE Xplore](#)

Research Interests and Technical Skills

Research Interests: Natural Language Processing, Machine Learning, Deep Learning, Generative AI, Fairness-Aware AI, Data Science

Programming: Python, Java, C++, SQL

Machine Learning Frameworks: TensorFlow, PyTorch, Scikit-learn

Research Tools: LaTeX, Git, GitHub, Google Colab, Jupyter Notebook

Selected Academic and Software Projects

- **World Networking System** 
 - Simulated a global computer network using RIP, NAT, DHCP, and DNS to support routing, addressing, and reliable communication.
- **DIU LeaderBoard** 
 - Designed a Flutter-based student leaderboard application using Firebase and Riverpod to manage rankings, authentication, and performance data.
- **Realistic Village Scene with Day-Night Cycle** 
 - Created a 2D OpenGL village animation in C++ with synchronized sun-moon movement and animated objects.
- **IoT Smart Parking System** 
 - Developed an IoT parking dashboard using ESP32, Firebase, RFID, ultrasonic sensors, IR sensors, LCD, and LEDs for real-time slot monitoring.

Certifications

- Computer Office Application — Bangladesh Technical Education Board, January–June 2018.
- National Skill Certificate-II — Graphic Design, 2021.
- Data Science Internship — CodeAlpha, 2025.
- Python with Machine Learning — Jagannath University, 2024–2025. Certificate: [EDGE-DSTS-106-2389-00003](#)

Scholarships and Awards

Best Paper Award, IEEE CS BDC Symposium 2024 — Jagannath University

Awarded by the IEEE Computer Society Bangladesh Chapter for the best paper presented at the IEEE CS BDC Symposium 2024, held on 22–23 November 2024. [Paper Link](#)

Dean's List Award, Fall 2023 — Daffodil International University

Conferred by the Faculty of Science and Information Technology for outstanding academic performance in the B.Sc. in CSE.

Merit-Based Tuition Scholarship — Daffodil International University

Awarded for 6 semesters during the B.Sc. in CSE programme: 3 semesters with a perfect SGPA of 4.00/4.00 and 3 semesters with SGPA above 3.90.

Hall Hero Award — Daffodil International University, YKSG-1

Recognizing dedication, discipline, and exemplary contributions to residential hall activities and student community life.

Co-curricular Activities and Societies

- Executive — Industry Collaboration, NLP & ML Lab, DIU.
- Volunteer — Red Crescent, UNICEF Bangladesh; Joint Editor — Rotary Club of Feni Town.
- Member — CPC, Cyber Security Club, SBSF, Daffodil Volunteering Society.

Language Proficiency

Bangla — Native || English — Professional Working Proficiency

References

Dr. Abdus Sattar

Associate Professor & Director (M.Sc.)
Dept. of Computer Science & Engineering
Daffodil International University (DIU),
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Dr. Md. Alamgir Kabir

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Dr. Arif Mahmud

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